Unit 1: Introduction to Quantum Theory and Technologies

The transition from classical to quantum physics, Fundamental principles explained conceptually: Superposition, Entanglement, Uncertainty Principle, Wave-particle duality, Classical vs Quantum mechanics – theoretical comparison, Quantum states and measurement: nature of observation,Overview of quantum systems: electrons, photons, atoms,The concept of quantization: discrete energy levels,Why quantum? Strategic, scientific, and technological significance,A snapshot of quantum technologies: Computing, Communication, and Sensing,National and global quantum missions: India's Quantum Mission, EU, USA, China

INTRODUCTION TO QUANTUM TECHNOLOGY AND ITS APPLICATIONS

The word **"quantum"** comes from the Latin word *quantus*, meaning **"how much."** In modern use, it has a few specific meanings depending on the field.

Quantum refers to the **smallest possible discrete unit** of any physical property.

For example:

A **quantum of light** is a **photon**.

A **quantum of energy** is the smallest packet of energy that can be emitted or absorbed.

So, in physics:

Quantum = a small, in divisible unit of a physical quantity

General use	A measurable amount or portion
Physics	Smallest possible unit of energy or matter
Quantum computing	A unit of quantum information (qubit)

Quantum technology is the principles of **quantum mechanics**—a fundamental theory in physics that describes nature at the smallest scales, such as atoms and subatomic particles—to develop new types of technologies with capabilities far beyond those of classical systems.

Introduction to Quantum Technology

What is Quantum Technology?

Quantum technology is based on key quantum mechanical properties such as

- **Superposition**: A quantum system can exist in multiple states at once until measured.
- **Entanglement**: Particles can be linked in such a way that the state of one instantly affects the other, no matter the distance.

What is Superposition?

The word "Superposition", which is a core concept in quantum mechanics and quantum technology.
Example:
A classical bit can be either: 0 or 1
A qubit in superposition can be:
•Both 0 and 1 at the same time, in a probabilistic blend, represented as:
$|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$
It is known as Dirac notation or bra-ket notation.
It can be read as
"ket psi equals alpha ket zero plus beta ket one"
where α and β are complex numbers that define the probabilities of being in each state.
•A "ket" represents a quantum state vector. For example, $|\psi\rangle$ denotes the state ψ of a quantum system. The ket is conceptually a column vector in a complex vector space called a Hilbert space. Specific qubit basis states like $|0\rangle$ and $|1\rangle$ represent the fundamental computational states in quantum computing.
•The "bra" is the dual vector (conjugate transpose) of the ket. If $|\psi\rangle$ is a column vector, then $\langle\psi|$ is the corresponding row vector with complex conjugated elements. Bras are used to form inner products and projections.

Dirac notation

- Dirac notation, also known as bra-ket notation, is a powerful and concise mathematical notation introduced by physicist Paul Dirac for representing **quantum states** and **operations in quantum mechanics**. It simplifies expressing **quantum states** and **computations** in a way that's especially useful in **quantum computing** and other **quantum technologies**.
- $|\psi\rangle$ denotes the state ψ of a quantum system

- **Inner Product $\langle\varphi|\psi\rangle$:**
  The inner product between two states $|\varphi\rangle$ and $|\psi\rangle$ is a complex number that gives the probability amplitude of state $|\psi\rangle$ collapsing to $|\varphi\rangle$ upon measurement. The squared magnitude $|\langle\varphi|\psi\rangle|^2$ gives the measurement probability.
- **Outer Product $|\psi\rangle\langle\varphi|$:**
  This represents an operator (or matrix) that acts on quantum states, useful for expressing quantum gates and measurements.

Quantum entanglement is a phenomenon in quantum physics where two or more particles become linked in such a way that the state of one particle instantly influences the state of the other(s), no matter how far apart they are even across vast distances.

When two particles (like photons, electrons, or atoms) interact in certain ways, their quantum states become *correlated*. After they separate, measuring one particle's state (say, its spin or polarization) will *instantly* determine the state of the other, regardless of the distance between them.

Photon: It refers to a particle of light or a quantum of electromagnetic radiation

EX:A photon travels at the speed of light. A photon is a quantum of light.

Matter: Matter is anything that has mass and takes up space (volume). Everything we see and touch is made of matter.

EX : Water, air, and wood are all forms of matter.

Atoms : An atom is the smallest unit of matter that still retains the properties of an element. Everything around us is made of atoms.

EX : All matter is made up of atoms.

Electron : An electron is a tiny particle with a negative electric charge, found in atoms, and responsible for electricity.

EX : Electrons revolve around the nucleus of an atom.

Main Branches of Quantum Technology

Quantum Computing

- Utilizes *qubits* (quantum bits) instead of classical bits (0 or 1).
- Capable of solving certain problems exponentially faster than classical computers.
- Key players: IBM, Google, Microsoft, IonQ, Rigetti.

Quantum Communication

- Ensures secure data transmission using quantum cryptography (e.g., Quantum Key Distribution - QKD).

Quantum Sensing and Metrology

- Provides ultra-precise measurements of time, gravity, magnetic fields, etc.
- Quantum sensors are used in navigation, geological exploration, and medical diagnostics.

Quantum Simulation

- Simulates complex quantum systems that are difficult to model with classical computers.
- Useful for understanding materials, drug discovery, and high-energy physics.

Applications of Quantum Technology

1. Cryptography and Cybersecurity

Quantum Key Distribution (QKD) allows unbreakable encryption methods.

Potential to secure national defence, financial transactions, and personal data.

2. Healthcare and Life Sciences

Quantum simulations for drug discovery and protein folding.

3. Financial Modelling

Quantum algorithms for portfolio optimization, risk analysis, and fraud detection.

4. Artificial Intelligence and Machine Learning

Quantum-enhanced machine learning (QML) offers faster training and data processing.

5. Climate Modelling and Material Science

Simulating atomic-scale processes helps develop new materials for batteries, solar panels, and carbon capture.

6. Navigation and Positioning

Quantum gyroscopes and accelerometers for GPS-free navigation (e.g., in submarines or spacecraft).

Current Challenges and Future Outlook

- **Scalability**: Building reliable, large-scale quantum systems remains a major hurdle.
- **Error Correction**: Quantum systems are highly susceptible to noise and decoherence.
- **Standardization**: Common protocols and infrastructure are still evolving.
- **Talent Gap**: A growing need for skilled professionals in quantum sciences.

The Transition from Classical to Quantum Physics

- Classical physics (Newtonian mechanics, Maxwell's equations) explains most macroscopic phenomena.
- However, it fails at atomic and subatomic scales.
- Problems like blackbody radiation, the photoelectric effect, and atomic spectra led to the development of quantum physics.
- Quantum theory emerged as a new framework to describe nature at these scales, where energy and matter behave differently.

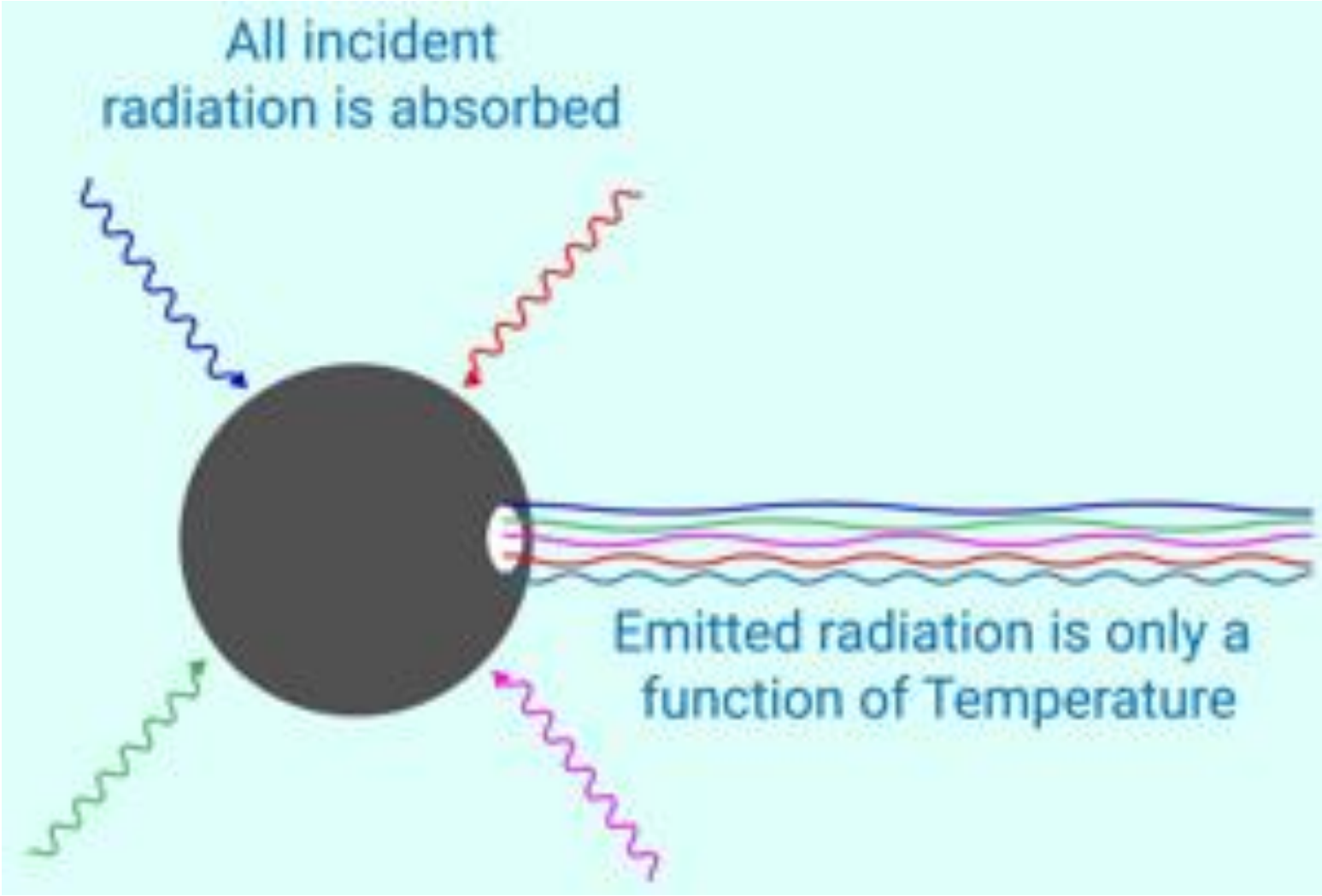
Example:

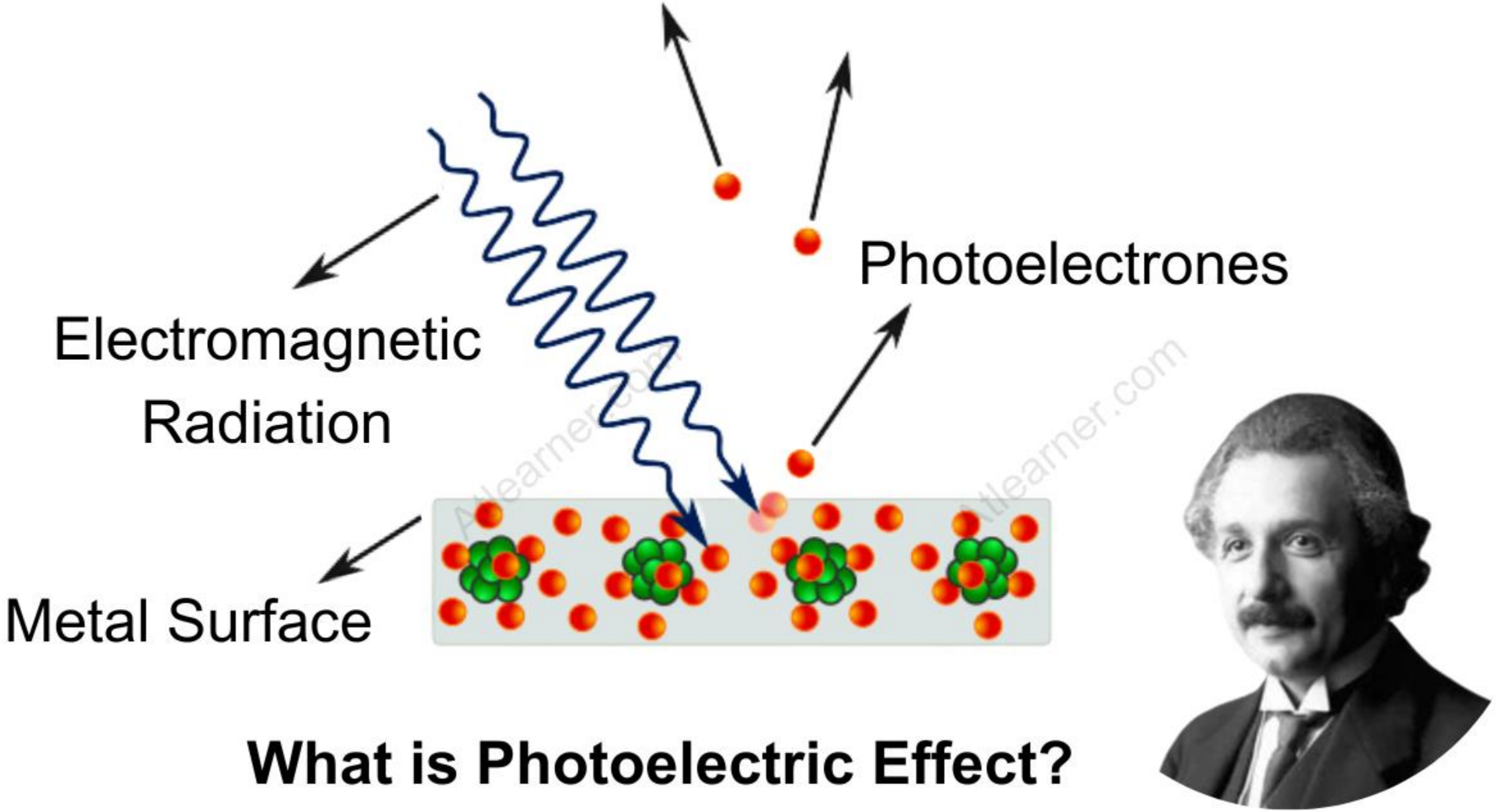
In the photoelectric effect, increasing the brightness of light (classical expectation) doesn't always eject electrons. Only light above a certain frequency does — this led to Einstein's quantum explanation involving photons.

1.Blackbody Radiation: Classical physics could not explain the observed spectrum of radiation emitted by a black body, which led Max Planck in 1900 to propose that electromagnetic energy is quantized, emitted in discrete packets called "quanta" or photons. This marked the birth of the quantum hypothesis.

2.Photoelectric Effect: Observed earlier but explained by Albert Einstein in 1905, the photoelectric effect showed that light must behave as discrete energy packets (photons) to eject electrons from a metal surface. The kinetic energy of these ejected electrons depended on the light frequency, not its intensity—contrary to classical wave theory predictions. Einstein's explanation provided strong evidence for quantization of light and helped establish quantum theory.

3.Atomic Spectra: Observations of discrete emission lines from atoms (like hydrogen) indicated that the energy levels of electrons are quantized. Bohr's model of the atom incorporated quantized electron orbits, further solidifying quantum ideas.


All incident
radiation is absorbed
Emitted radiation is only a
function of Temperature


Photoelectrones
Electromagnetic
Radiation
Metal Surface
What is Photoelectric Effect?

Fundamental Principles Explained Conceptually

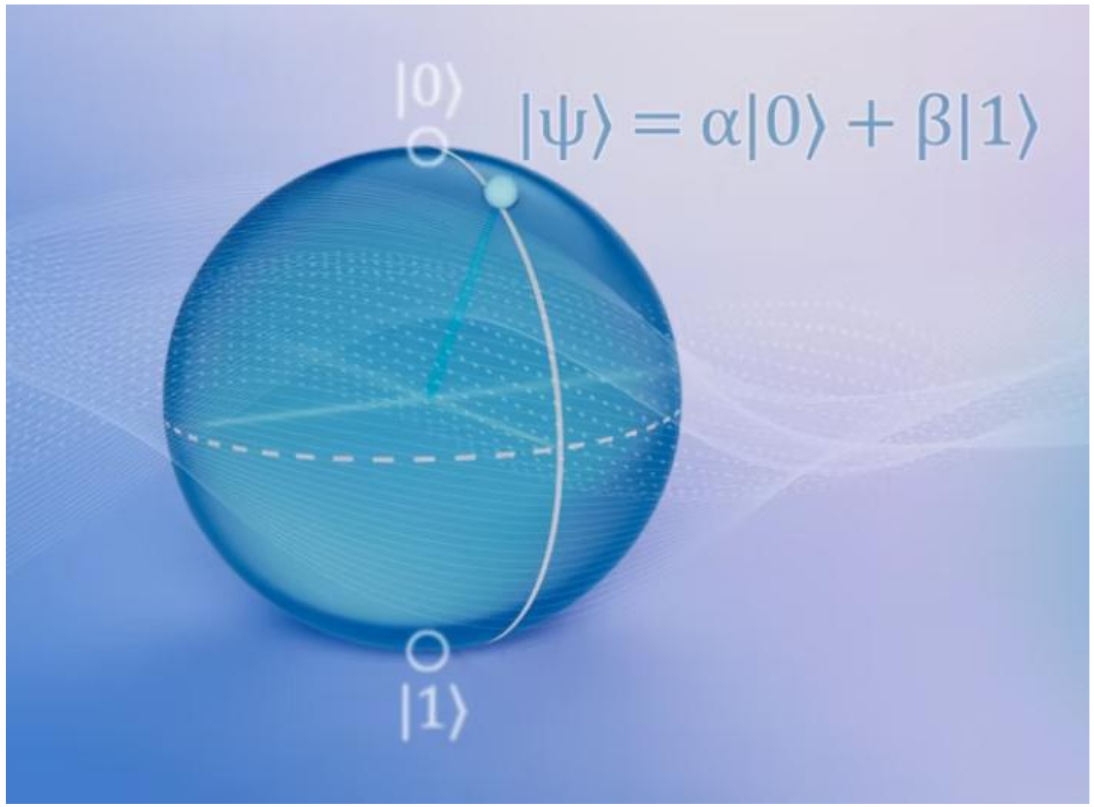
A . Superposition:-

A quantum particle can exist in multiple states at the same time. This is called superposition.

Example:

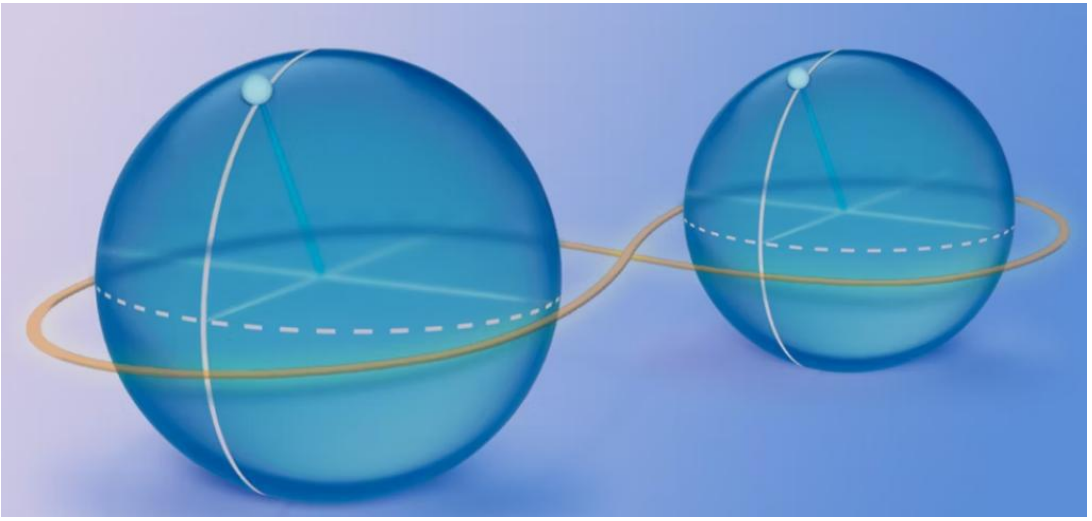
A qubit (quantum bit) in a quantum computer can be in both $|0\rangle$ and $|1\rangle$ states at once.
$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are complex numbers

with $\alpha^2 + \beta^2 = 1$.



B . Entanglement:-

- **Two particles become entangled if the state of one instantly affects the state of the other, no matter the distance between them.**
- **Example:**
- **If two photons are entangled and one is measured to be vertically polarized, the other will instantly be horizontally polarized, even if it's kilometers away.**



Uncertainty Principle:-

Heisenberg's principle says **we cannot simultaneously know** a particle's position and momentum exactly.

Example:

If we know an electron’s position very precisely (e.g., inside a tiny box), we cannot know its momentum — it might be moving rapidly or slowly, and we cannot predict both exactly.

D . Wave-Particle Duality:-

Quantum objects can behave both like **particles and waves**, depending on how we observe them.

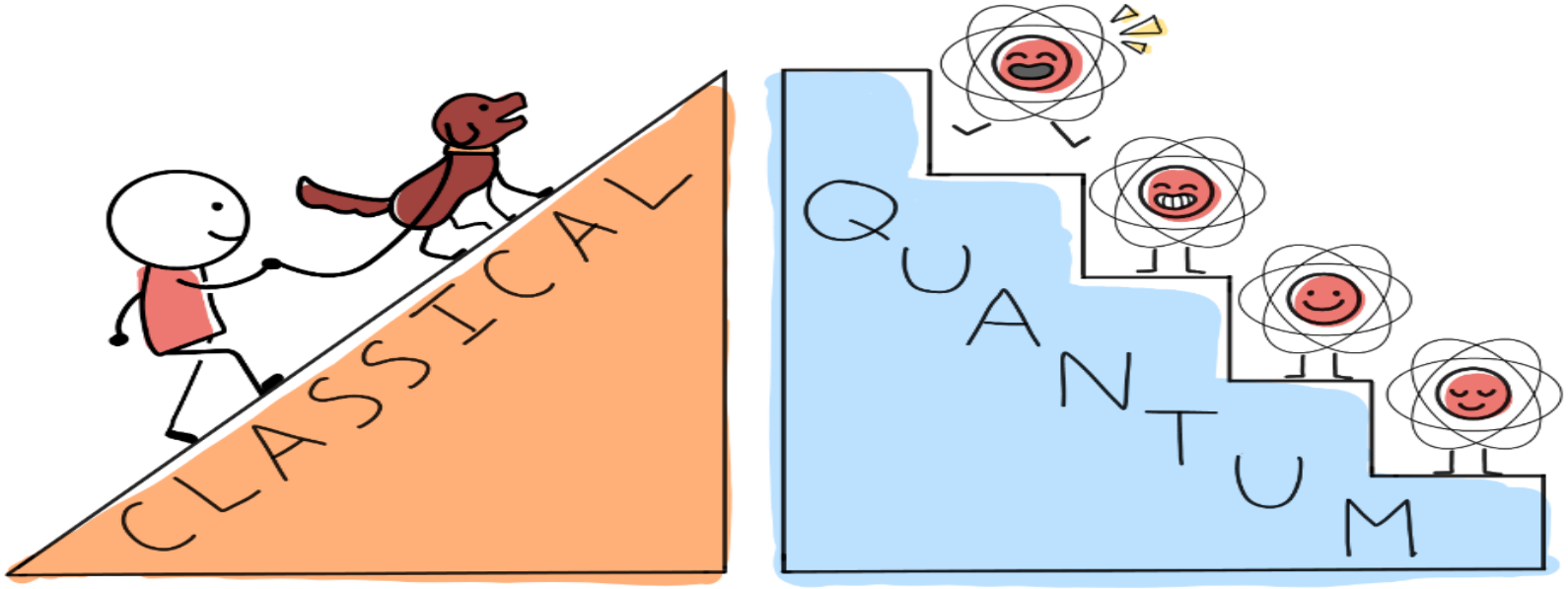
Example:

In the **double-slit experiment**, electrons fired one at a time still create an **interference pattern** like waves, until we observe which slit they pass through—then they behave like particles.

Classical vs Quantum mechanics – theoretical comparison

Key points	Classical computing	Quantum computing
Basis of computing	Large scale multipurpose computer based on classical physics	High speed computer based on quantum physics
Information storage	Bit based using voltage or charge	Quantum bit based using electron spin or polarisation
Bit values	Bits having a value either 0 or 1 can have a single value at any instant	Qubits have a value of 0, 1 or sometimes linear combination of both (a property known as superposition)
Number of possible states	The number of possible states is 2 which is either 0 or 1	The number of possible states is infinite since it can hold combination of 0 or 1 along with some complex information
Output	Deterministic (repetition of computation on the same input gives the same output	Probabilistic (repetition of computation on superposed states gives probabilistic answer)
Gates used for processing	Logic gates (AND, OR, NOT, etc)	Quantum gates (X, Y, Z, H, CNOT etc.)
Operations	Operations use Boolean Algebra	Operations use linear algebra and are represented with unitary matrices

Classical vs Quantum mechanics



Classical mechanics treats motion and energy as continuous, and systems evolve along smooth trajectories in space and time. Quantum mechanics, however, reveals that energy is quantized—only specific, discrete energy levels are allowed

Quantum states and measurement

What is a Quantum State?

A quantum state describes the complete information about a quantum system (like an electron, photon, or atom).

It is usually represented as a wavefunction (Ψ) in quantum mechanics.

It tells you everything that can be known about a particle—such as its probability of position, momentum, spin, etc.

Mathematically:

A quantum state is a vector in a Hilbert space.

In Dirac notation, it's written as:

$|\psi\rangle$

Example:

A photon polarized horizontally: $|H\rangle$

A qubit in superposition:

$|\psi\rangle=\alpha|0\rangle+\beta|1\rangle$ where α and β are complex numbers and

$|\alpha|^2+|\beta|^2=1$

What is Quantum Measurement?

Measurement in quantum mechanics is not like classical measurement.

It has special rules:

1.Measurement collapses the state:

- Before measurement, the system exists in a superposition (multiple possibilities).
- When you measure it, the wavefunction "collapses" into one definite outcome.
- After that, the system behaves classically (until disturbed again).

2.Outcome is probabilistic:

- The result of the measurement is not deterministic.
- Each possible result has a probability, given by the square of the amplitude: $P=|\alpha|^2$

Die-in-a-box analogy	Quantum mechanics term
Die rolling unseen inside the box	Superposition — system exists in a combination of possible states
Possible die faces (1–6)	Eigenvalues — possible measurement results
Opening the box to look	Measurement — interacting with the system to get a result
Die suddenly shows one fixed number	Wavefunction collapse — system picks one definite state
Rules of the die roll (fair/unfair die)	Probability amplitudes — wavefunction determines chance of each outcome
Seeing the same number if checked again immediately	Repeatability postulate — once measured, the state stays the same until disturbed
Shaking the box again before looking	New state preparation — changes the superposition

Example:

Spin Measurement

An electron's spin can be up or down along a direction:

Before measurement:

$|\psi\rangle = \frac{1}{\sqrt{2}}\,(|\uparrow\rangle + |\downarrow\rangle)$

After measurement along z-axis, it becomes either:

$|\uparrow\rangle$ **with 50% chance**

$|\downarrow\rangle$ **with 50% chance**

Momentum

Momentum is the quantity of motion an object has. It depends on how massive the object is

Momentum=mass × velocity (Or)

p=mv Where:

p = momentum (in kg · m/s)

m = mass (in kilograms)

v = velocity (in meters per second)

and how fast it is moving.

Nature of Observation in Quantum Mechanics

In quantum physics, the nature of observation is *fundamentally different* from classical science. It's not just about "looking" — observation changes the system being observed.

What Does "Observation" Mean?

In quantum theory, observation (or measurement) refers to interacting with a quantum system to extract information — like the position, momentum, or spin of a particle.

But unlike in classical physics, this interaction affects the system in a fundamental way.

Example: Double-Slit Experiment

When electrons are fired through two slits:

- **If not observed, they create an interference pattern — like waves.**
- **If you observe which slit they go through, the interference disappears, and they act like particles.**

The Concept of Quantization: Discrete Energy Levels

What is Quantization?

In quantum physics, **quantization** means that certain physical quantities (like energy, charge, angular momentum) can take **only specific, discrete values**, not a continuous range.

Example: Hydrogen Atom

In the hydrogen atom, the electron **can't orbit at any distance** from the nucleus.
Instead, it occupies **quantized energy levels** (n = 1, 2, 3...).
When the electron **jumps between levels**, it **absorbs or emits** energy as **photons**.
$E_n = -13.6\ eV/n^2$
This explains the **line spectra** of elements (e.g., Balmer series for hydrogen).

More Examples:

System	Quantized Property
Electron in atom	Energy levels
Photons in a laser	Discrete energy packets (quanta)
Particle in a box	Quantized momentum & energy
Light	Comes in photons with E=hf

Why Quantum Technologies ?:-

Highlighting its **strategic**, **scientific**, and **technological significance**, **with examples** for each

1. Scientific Significance

Quantum mechanics fundamentally changed how we understand the universe.

Key Concepts:

Superposition – A quantum system can be in multiple states at once.

Entanglement – Two particles can be connected over long distances.

Uncertainty Principle – we can't precisely know both position and momentum.

Example: The **photoelectric effect** is the **emission of electrons** from the surface of a metal when **light of a certain frequency (or higher)** falls on it.

These emitted electrons are called **photoelectrons**.

Photoelectric Effect: Explained by Einstein using quantum theory. Classical physics failed to explain why light below a certain frequency doesn't eject electrons, but quantum theory did—earning Einstein the Nobel Prize.

Atomic Model: Niels Bohr's model of the atom couldn't work without quantum energy levels.

2. Technological Significance

Quantum mechanics is being **applied to build radically new technologies**.

Quantum Technologies:

Quantum Computing: Solves problems beyond classical computers.

Quantum Communication: Enables hack-proof data transfer.

Quantum Sensing: Allows ultra-precise measurement.

Examples:

Quantum Computers: Google's *Sycamore* quantum processor solved a problem in 200 seconds that would take a classical supercomputer 10,000 years.

Quantum Key Distribution (QKD): China's *Micius satellite* transmitted entangled photons over 1,200 km for secure communication.

Quantum Sensors: Used for underground mineral detection and GPS-independent navigation (by detecting gravitational variations).

3. Strategic Significance

Quantum tech affects **national security, cybersecurity, and economic power**.

Key Strategic Areas:

Cybersecurity: Quantum computers threaten existing encryption (e.g., RSA, ECC).

Defense: Quantum radars can detect stealth objects; quantum navigation works without GPS.

Economic Superiority: Quantum innovation drives future industrial competitiveness.

Examples:

USA: National Quantum Initiative invested $1.2+ billion; tech giants like IBM and Microsoft lead in quantum R&D.

China: Leads in quantum communication and has built the world's longest quantum network.

India: Launched the **National Quantum Mission** in 2023 with ₹6003 crore to develop quantum computing, communication, and sensing.

Military Use: Quantum magnetometers and gyroscopes are being tested for submarines and aircraft to operate without GPS.

Aspect	Significance	Example
Scientific	Explains atomic and subatomic behavior; foundation of modern physics	Photoelectric effect, Bohr's atom model
Technological	Enables new tools in computing, sensing, and communication	Google Sycamore, Micius satellite, quantum gyroscopes
Strategic	Determines future cybersecurity, military tech, and economic leadership	National Quantum Missions (US, China, India), Quantum encryption

A snapshot of quantum technologies: Computing, Communication, and Sensing

- **Quantum technologies**, focusing on the three major areas:
- **Computing**
- **Communication**
- **Sensing.**
- Quantum technologies leverage the principles of quantum mechanics — like **superposition**, **entanglement**, and **quantum tunnelling.**

1. Quantum Computing

Definition:

Quantum computing uses **qubits** instead of classical bits. A qubit can be in a superposition of 0 and 1, allowing quantum computers to perform parallel computations. It is all bout qubits, quantum gates, quantum circuits, and algorithms such as Grover's and Shor's that outperform classical solutions

Key Principles:

- Superposition
- Entanglement
- Quantum interference

Applications:

Cryptography: Breaking classical encryption algorithms (like RSA) using **Shor's algorithm**.

Optimization: Solving complex problems in logistics and finance.

Drug discovery: Simulating molecular interactions quickly and accurately (e.g., by companies like **IBM** and **Google Quantum AI**).

Machine learning: Speeding up learning algorithms with quantum-enhanced data processing.

Example:

2. Quantum Communication

Definition:
Quantum communication uses quantum states (like the polarization of photons) to transfer information securely.

Key Principle:

Quantum Key Distribution (QKD) using **entanglement** ensures that any eavesdropping attempt disturbs the system, making it detectable.

Applications:

Secure communication networks (unhackable).

Satellite-based quantum internet.

Quantum teleportation of information (not matter).

Example:

China's Micius Satellite: World's first quantum communication satellite. Demonstrated secure **quantum key exchange** between two ground stations over 1,000+ km.

India's Quantum Key Distribution (QKD) test between DRDO and IIT Delhi in 2022.

3. Quantum Sensing

Definition:

Quantum sensors exploit quantum properties to detect and measure extremely small changes in physical quantities like time, magnetic fields, gravity, or acceleration.

Key Principle:

Use of superposition and entanglement in atoms, ions, or photons to enhance measurement sensitivity.

Applications:

Navigation without GPS using quantum gyroscopes.

Medical imaging with ultra-high resolution (e.g., using quantum-enhanced MRI).

Geological exploration: Detecting underground structures with quantum gravimeters.

Atomic clocks: Most precise timekeeping devices for GPS and scientific research.

Example:

Quantum Gravimeters developed by companies like **Q-CTRL** and **Muquans** for detecting underground voids and oil reservoirs.

Optically pumped magnetometers used in brain imaging (MEG scans) to detect tiny magnetic fields produced by neural activity.

National and global quantum missions: India's Quantum Mission, EU, USA, China

National and Global Quantum Missions — focusing on **India, the European Union (EU), the USA, and China**, along with major goals and examples from each.

National and Global Quantum Missions

Quantum technologies are becoming strategically important worldwide. Countries are investing heavily to lead in **Quantum Computing**, **Communication**, and **Sensing**, due to their potential to revolutionize national security, healthcare, finance, and more.

IN India's National Quantum Mission (NQM)

Launched: Approved by the Union Cabinet in **April 2023**

(by DST – Department of Science and Technology)

Budget: ₹6003.65 Crores (~$730 million) for 8 years

Goals:

Develop **50–100 qubit quantum computers** by 2031

Establish **quantum communication** across 2000 km of optical fiber

Create **national quantum labs**, innovation hubs, and training programs

Build skilled manpower and promote **indigenous technology**

Key Projects:

Quantum Key Distribution (QKD) trials (e.g., DRDO-IIT Delhi collaboration)

Satellite-based secure communication

Quantum Simulation and Quantum Materials research hubs

Example:

India's QKD demonstration between Prayagraj and Vindhyachal (100+ km) by C-DOT for secure power grid communication.

EU European Union – Quantum Flagship Program

Launched: 2018
Budget: €1 billion (2018–2028)
Framework: Under Horizon Europe

Goals:

- Strengthen European leadership in **quantum tech**
- Commercialize quantum computing and sensing applications
- Develop **quantum internet infrastructure** for secure communication

Initiatives:

OpenSuperQ: Building a quantum computer with 100+ qubits

Quantum Internet Alliance (QIA): Developing a pan-European quantum network

QuIC (Quantum Industry Consortium): Connecting academia and startups

Example:
The **Quantum Internet Alliance** is working on connecting multiple European quantum labs through entangled networks.

US United States – National Quantum Initiative (NQI)

Launched: 2018, under the **National Quantum Initiative Act**
Budget: Over $1.2 billion over five years (initially)

Goals:

- Build **large-scale quantum computers**
- Enhance **quantum workforce development**
- Strengthen collaborations between government, academia, and industry

Institutions Involved:

DOE (Department of Energy)

NIST (National Institute of Standards and Technology)

NSF (National Science Foundation)

Major Centers:

- **Q-NEXT** (Argonne National Lab)
- **Quantum Leap Challenge Institutes**
- **IBM, Google, Microsoft** all running commercial quantum programs

Example:

In 2019, Google AI (USA) achieved **quantum supremacy** using its 53-qubit **Sycamore** processor.

CN China – Quantum Leap Program

Launched: Around 2016 under **"Made in China 2025"** strategy
Investment: Over $10 billion (e.g., National Laboratory for Quantum Information Sciences in Hefei)

Goals:

- Lead in **quantum communication, computing, and sensing**
- Develop **quantum satellites**, secure networks, and large-scale quantum computing

Achievements:

- **Micius Satellite**: First quantum satellite enabling entanglement-based QKD over 1200 km
- World's first **quantum-encrypted communication** between Beijing and Shanghai
- Progress in **photonic quantum computing** (e.g., Jiuzhang computer)

Example:

China's **Jiuzhang** quantum computer performed **Gaussian boson sampling** faster than any classical supercomputer — showing **quantum advantage** in a photonic system.

END UNIT-1